Khuzaima Hassan

Software Engineer |ML Systems |Python Developer

khuzaimahassan52@gmail.com | 923213937589 | khuzaima-hassan-2004mkh | github.com/KhuzaimaHassan | Karachi, Pakistan

PROFILE

Data Science undergraduate with strong problem-solving skills and hands-on experience in machine learning, LLM-powered applications, and data-driven systems. Skilled in Python, SQL, Streamlit, and model evaluation pipelines. Passionate about building intelligent tools that simplify data exploration and decision-making.

PROFESSIONAL EXPERIENCE

10Pearls, Data Science Intern 12/2025—2/2026

- Developed an ML-based Air Quality Index prediction system using Python for data preprocessing, feature engineering, and model training.
- Built an interactive application to visualize AQI predictions and environmental data insights.

Japanese Language & IT Internship (Plus W × OEC), *Internee* 01/2025 – 12/2025 | Remote

- Worked on real-world tech assignments.
- Gained practical experience with IT workflows, documentation, and coding tasks.
- Collaborated on internal prototypes using Python and data visualization.

PROJECTS

Risk Scoring Platform, End-to-End ML System

- Built an end-to-end fraud risk scoring system using Python, PostgreSQL, and FastAPI with feature engineering, and model training.
- Implemented model versioning and monitoring with statistical drift detection to track prediction distributions and system performance.

AI Academic Mentor, *Data Science Hackathon*

- Integrated Groq API to generate personalized student academic reports based on performance.
- Built a streamlit UI to showcase student data and report generation.

Insight Agent, *LLM Integration & Streamlit App*

- Integrated Gemini API (LLM) to answer queries over uploaded datasets in real time.
- Built a Streamlit dashboard to showcase dataset summary, visualizations, and interactive Q&A.

Real-Time Fraud Detection System, *Machine Learning*

- Designed balanced datasets using SMOTE, threshold tuning, and multiple ML models.
- Prepared evaluation metrics and Streamlit dashboard integration.

Real Estate Management System, *MySQL Database*

- Designed normalized relational schema and CRUD operations for property listings.
- Implemented stored procedures and query optimizations to support common use cases (search, filter, transactions).

EDUCATION

B.S. Data Science, *NED University of Engineering & Technology* 09/2022 – Present

- CGPA: 3.86
- Relevant coursework: Machine Learning, Database Systems, Algorithms, Data Visualization

Intermediate, *Adamjee Govt. Science College* 08/2020 – 08/2022

- Secured A-1 Grade with 93.51% marks.

Matriculation, *Usman Public School System* 04/2018 – 04/2020

- Awarded consecutive shields for achieving A+ grades in all subjects.

TECHNICAL SKILLS

Python
pandas, numpy, scikit-learn, matplotlib, seaborn

ML & AI
Classification models, regression, model evaluation, cross-validation, pipeline design, LLM integrations

Visualization
Power BI dashboards, modeling

SQL
Database design, Query optimization, Joins, Indexing, CRUD operations

Tools
FastAPI, Streamlit, Hugging Face, Jupyter Notebook, Git/GitHub

Networking
Huawei eNSP (RIP, VLAN, DHCP, Eth-Trunk/LACP)

CERTIFICATION

Machine Learning Scientist in Python
DataCamp

Python for Data Science, AI & Development
Coursera

Huawei Certified ICT Associate (HCIA)
Datacom

Data Analyst in SQL
DataCamp

Power BI Fundamentals
DataCamp

LANGUAGES

English: Professional working proficiency

Urdu: Native or bilingual

German: Elementary proficiency

Japanese: Elementary proficiency